MSc. Project Report: Survey on Equations over Finite Fields

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Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^{\infty} e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^{\top}C_fX$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i[d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Finite Fields

§ 1.1 Field Extensions

We build up the theory of field extensions following Morandi [Mor96].

The study of field theory is primarily the study of field extensions. The classical problems of ruler-and-compass constructions and the solvability of polynomial equations by radicals were settled by analyzing appropriate field extensions. The central idea is to associate a group—the *Galois group*—to a field extension, converting field-theoretic questions into group-theoretic ones. Since the Galois group of a finite extension is finite, the rich structure theory of finite groups becomes available. In this section we build the foundational vocabulary: degrees, algebraic elements, minimal polynomials, and the key structural results about simple and algebraic extensions.

1.1.1 Degrees and Generated Subfields

Let $F \subseteq K$ are fields. Then K is called a *field extension* of F. We denote it by K/F. Here F is called the *base field*. We can think of K as a vector space over F where the addition of two vectors follows the addition operation of K and the multiplication of any $\alpha \in K$ and $c \in F$ follows the multiplication operation of K, $c \cdot \alpha = c\alpha$. Then we denote the dimension of K over F by

$$[K\colon F] \coloneqq \dim_F(K)$$

This dimension is called the *degree of field extension* of F or simply *degree of* $[K\colon F]$. If $[K\colon F] < \infty$ we call it by a special name, *finite extension*. Finite extensions of finite fields are of very important to us and this report we will work on finite fields and their finite extensions. Let $\alpha \in K$. Then we denote $F(\alpha)$ to be the field generated by F and α following the multiplication and addition operations of K. We will define what it means to be the field generated by F and α in Definition 1.1.1. It is very easy to sea $F(\alpha)$ is also a field, and it contains F but also contained in K.

Let F be any field and suppose X be a formal variable. Then $F[X]$ denotes the polynomial ring over F. We define the rational function field $F(X)$ to be the *fraction ring* of $F[X]$ which consists of all quotients f/g where $f, g \in F[X]$ with $g(X) \neq 0$. Similarly, we can define the polynomial ring $F[X_1, \ldots, X_n]$ over multiple formal variables and their fraction field $F(X_1, \ldots, X_n)$.

Look at the following example. $\mathbb{R} \subseteq \mathbb{C}$ are both fields. But $\mathbb{C}$ is a 2-dimensional vector space over $\mathbb{R}$ as we can take the basis $\{1, i\}$. So $\mathbb{C}/\mathbb{R}$ is a finite extension of degree 2. But $\mathbb{Q}/\mathbb{R}$ is an infinite field extension. On the other hand let $\alpha \in \mathbb{R}$. Then consider the field $\mathbb{Q}(\alpha)$. Now degree of $[\mathbb{Q}(\alpha)\colon \mathbb{Q}]$ depends on what kind of element α is. If $\alpha = \sqrt{2}$ then $[\mathbb{Q}(\sqrt{2})\colon \mathbb{Q}] = 2$ but if $\alpha = \pi$ then $[\mathbb{Q}(\pi)\colon \mathbb{Q}] = \infty$.

Definition 1.1.1: Generated Subfields

Let K/F be a field extension and $S \subseteq K$.

(i) The ring generated by F and S, *written $F[S]$, is the intersection of all subrings of K containing F and S.*

(ii) The field generated by F and S, *written $F(S)$, is the intersection of all subfields of K containing F and S.*

If $S = \{a_1, \dots, a_n\}$ is finite we write $F[a_1, \dots, a_n]$ and $F(a_1, \dots, a_n)$. We call $F(\alpha_1, \dots, \alpha_n)$ finitely generated *field.*

From this definition it easily follows that for any $S \subseteq K$, $F(S)$ is the smallest subfield of K containing F and α.

Let K/F be a field extension. For any $\alpha \in K$ consider the map $\mathrm{ev}_\alpha : F[X] \to K$ defined by $f(X) \mapsto f(\alpha)$. Hence, ev_α is a ring homomorphism.

Observation 1.1. *For any $\alpha \in K$, $\mathrm{Im}(\mathrm{ev}_\alpha)$ is a subring of K and ev_α is F-vector space homomorphism.*

Now we can use this map ev_α to give descriptions of the elements of $F[\alpha]$ and $F(\alpha)$.

Lemma 1.1.1

Let K be a field extension of F and let $a \in K$. Then

$$F[a] = \{f(a) : f(x) \in F[x]\} \quad \text{and} \quad F(a) = \left\{\frac{f(a)}{g(a)} : f, g \in F[x],\ g(a) \neq 0\right\}.$$

Moreover, $F(a)$ is the quotient field of $F[a]$.

Proof: The evaluation map ev_a is a ring homomorphism with image $\{f(a) : f \in F[x]\}$/ So this set is a subring of K. Any subring of K containing F and a must contain $f(a)$, for all $f \in F[X]$. Therefore, $F[a] = \{f(a) : f \in F[x]\}$. The fraction field of $F[a]$ is $\{f(a)/g(a) \colon g(a) \neq 0\}$; it is contained in any subfield of K containing $F[a]$, hence equals $F(a)$. ■

We call this procedure *adjoining* α to F. Using similar arguments we can describe the ring $F[a_1, \dots, a_n]$ and the field $F(a_1, \dots, a_n)$ where K/F is a field extension and $a_1, \dots, a_n \in K$. So we have the following theorem.

Theorem 1.1.2

Let K be a field extension of F and let $a_1, \dots, a_n \in K$. Then

$$F[a_1, \dots, a_n] = \{f(a_1, \dots, a_n) : f \in F[x_1, \dots, x_n]\},$$
$$F(a_1, \dots, a_n) = \left\{\frac{f(a_1, \dots, a_n)}{g(a_1, \dots, a_n)} : f, g \in F[x_1, \dots, x_n],\ g(a_1, \dots, a_n) \neq 0\right\}.$$

Moreover, $F(a_1, \dots, a_n)$ is the quotient field of $F[a_1, \dots, a_n]$.

Now we can also define fields generated by adjoining arbitrary number of elements. In this case we can give the description of the new field in terms of all finite extension of fields created by adjoining finitely many elements.

Theorem 1.1.3

Let K be a field extension of F and let $S \subseteq K$ be any subset. If $\alpha \in F(S)$, then $\alpha \in F(a_1, \dots, a_n)$ for some $a_1, \dots, a_n \in S$. Therefore,

$$F(S) = \bigcup \{F(a_1, \dots, a_n) \colon a_1, \dots, a_n \in S\},$$

where the union is over all finite subsets of S.

Proof: Let $U := \cup\{F(a_1,\dots,a_n) : a_1,\dots,a_n \in S\}$. Each $F(a_1,\dots,a_n)$ contains F and lies in $F(S)$ where $a_1,\dots,a_n \in S$. So the union U satisfies $U \subseteq F(S)$. The set U contains F and S. If $\alpha \in F(a_1,\dots,a_n)$ and $\beta \in F(b_1,\dots,b_m)$, then $\alpha \pm \beta$, $\alpha\beta$, α/β (if $\beta \neq 0$) are all in $F(a_1,\dots,a_n,b_1,\dots,b_m) \subseteq U$. Therefore, U is a field. Being the smallest subfield containing F and S, it equals $F(S)$. ■

1.1.2 Algebraic Elements and Minimal Polynomials

Definition 1.1.2: Algebraic and Transcendental Elements

Let K be a field extension of F. An element $\alpha \in K$ is algebraic over F *if there is a nonzero polynomial $f(X) \in F[X]$ with $f(\alpha) = 0$. If α is not algebraic over F, then α is called* transcendental over F. *If every element of K is algebraic over F, then K is said to be* algebraic over F, *and K/F is called an* algebraic extension.

Hermite (1873) proved that e is transcendental over $\mathbb{Q}$ and Lindemann (1882) proved the same for π. On the other hand, $\sqrt[n]{r} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ for any $r \in \mathbb{Q}$ (root of $x^n - r$), as is $i = \sqrt{-1}$ (root of $x^2 + 1$).

Definition 1.1.3: Minimal Polynomial

If α is algebraic over a field F, the minimal polynomial *of α over F is the monic polynomial $p(X)$ of least degree in $F[X]$ for which $p(\alpha) = 0$; it is denoted $\min(F,\alpha)$. Equivalently, $\min(F,\alpha)$ is the monic generator of the kernel of the evaluation homomorphism ev_α.*

The minimal polynomial depends on the base field. For instance, $\min(\mathbb{Q}, i) = x^2 + 1$ while $\min(\mathbb{C}, i) = x - i$, since $i \in \mathbb{C}$ is already in the base field.

The minimal polynomial of an element and the degree of a field extension are two of the most basic tools we shall use. The following proposition gives a relation between these objects.

Theorem 1.1.4

Let K/F be a field extension and let $\alpha \in K$ be algebraic over F. If $p(X) = \min(F,\alpha)$, then:

(i) *$p(X)$ is irreducible over F.*

(ii) *If $g(X) \in F[X]$, then $g(\alpha) = 0$ if and only if $p(X) \mid g(X)$.*

(iii) *If $n = \deg(\min(F,\alpha))$, then the elements $1, \alpha, \dots, \alpha^{n-1}$ form a basis for $F(\alpha)$ over F. So $[F(\alpha) : F] = \deg(\min(F,\alpha)) < \infty$. Moreover, $F(\alpha) = F[\alpha]$.*

(iv) *$F(\alpha) \cong F[X] \,/\, \langle p(X) \rangle$.*

Proof: The *first isomorphism theorem* gives $F[\alpha] \cong F[X] \,/\, \langle p(X) \rangle$. Since $F[\alpha] \subseteq K$ is an integral domain, $\langle p(X) \rangle$ is a prime ideal. Now in a PID every nonzero prime is maximal, so $F[X] \,/\, \langle p(X) \rangle$ is a field and $F[\alpha] = F(\alpha)$. This gives (i) and (iv). Part (ii) follows from $\ker(\mathrm{ev}_\alpha) = \langle p(X) \rangle$.

For (iii): any $b \in F(\alpha) = F[\alpha]$ satisfies $b = g(\alpha)$. Dividing gives $g = qp + r$ with $\deg r < n$ and $q, r \in F[X]$. So $b = r(\alpha)$ lies in the span of $\{1, \alpha, \dots, \alpha^{n-1}\}$. Therefore, the set $\{1, \alpha, \dots, \alpha^{n-1}\}$ spans $F(\alpha)$ as a vector space. If $a_0 + a_1 \cdot \alpha + \cdots + a_{n-1} \cdot \alpha^{n-1} = 0$ where $a_0, \dots, a_{n-1} \in F$ then the polynomial $f(X) = \sum_{i=0}^{n-1} a_i X^i$ is in $\ker(\mathrm{ev}_\alpha)$. Hence, either $f \equiv 0$ or $p \mid f$ by part (ii). But $\deg(f) < n$. So $f \equiv 0$. Thus, $1, \alpha, \dots, \alpha^{n-1}$ forms a basis of $F(\alpha)$ over F. ■

The part (iii) of above theorem gives us a basis of $F(\alpha)$ over F using only α. So we get the following observation:

Observation 1.2. *If K/F is a field extension and $\alpha \in K$ is algebraic over F where $p = \min(F, \alpha)$ then $[F(\alpha) : F] = \deg(p)$.*

Notice that if $\alpha, \beta \in K$ are both algebraic over F and $\min(F, \alpha) = \min(F, \beta) = p$ i.e., their minimal polynomials are same then by the above theorem

$$F(\alpha) \cong {F[X]}\Big/{\langle p \rangle} \cong F(\beta)$$

Therefore we have the observation:

Observation 1.3. *If K/F is a field extension and $\alpha, \beta \in K$ are both algebraic over F, and they have the same minimal polynomials then $F(\alpha) \cong F(\beta)$ with the isomorphism $\varphi : \alpha \mapsto \beta$.*

1.1.3 Finite and Algebraic Extensions

Lemma 1.1.5

If $[K : F] < \infty$, then K is algebraic and finitely generated over F.

Proof: Let $\{\alpha_1, \ldots, \alpha_n\}$ be an F-basis of K. Then for any element $\alpha \in K$, there exists $a_1, \ldots, a_n$ such that $\alpha = a_1 \cdot \alpha_1 + \cdots + a_n \cdot \alpha_n$. So certainly we have $K = F(\alpha_1, \ldots, \alpha_n)$ and hence K is finitely generated over F. Since $[K : F] = n$, for any $\alpha \in K$, the $n+1$ elements $1, \alpha, \ldots, \alpha^n$ span only an n-dimensional space. So they are F-linearly dependent. Thus, there exists $\beta_0, \ldots, \beta_n \in F$ such that $\sum_{i=0}^{n} \beta_i \cdot \alpha^i = 0$. So we get a polynomial $f(X) = \sum_{i=0}^{n} \beta_i \cdot X^i$ and $f(\alpha) = 0$. Therefore, α is algebraic over F. ■

Let K/F is a finite extension and $K = F(\alpha_1, \ldots, \alpha_n)$. Then we can break up the extension K/F into collection of subextensions. Let $L_i = F(\alpha_1, \ldots, \alpha_i)$. Set $L_0 = F$. Then basically $L_{i+1} = L_i(\alpha_{i+1})$. And we have

$$F = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_{n-1} \subseteq L_n = K$$

We will show a theorem which relates the degree of the extension K/F and each subextensions.

Theorem 1.1.6

Let $F \subseteq L \subseteq K$ be fields. Then $[K : F] = [K : L] \cdot [L : F]$.

Proof: Let $\{a_i\}_{i \in I}$ be an F-basis of L and $\{b_j\}_{j \in J}$ an L-basis of K. We show $\{a_i \cdot b_j\}_{I \times J}$ is an F-basis of K. For any $\gamma \in K$, write $\gamma = \sum_j \alpha_j b_j$ with $\alpha_j \in L$. Now each $\alpha_j = \sum_i \beta_{ij} a_i$ with $\beta_{ij} \in F$; then $\gamma = \sum_{i,j} \beta_{ij} a_i b_j$. Therefore, the set $\{a_i \cdot b_j\}_{I \times J}$ spans K over F.

If $\sum_{i,j} \beta_{ij} a_i b_j = 0$ for some non-trivial linear combinations where $\beta_{i,j} \in F$ for all $i \in I$, $j \in J$. Now

$$\sum_{i,j} \beta_{ij} a_i b_j = \sum_j \Big(\sum_i \beta_{ij} a_i\Big) b_j = 0.$$

Since $\{b_j\}_{j \in J}$ is a basis of K over L, each $\sum_i \beta_{ij} a_i = 0$. And by F-independence of $\{a_i\}$, each $\beta_{ij} = 0$. So $\{a_i \cdot b_j\}_{I \times J}$ form a F-basis of K. Therefore,

$$[K : F] = |\{a_i \cdot b_j\}_{I \times J}| = |\{a_i : i \in I\}| \cdot |\{b_j : j \in J\}| = [K : L] \cdot [L : F]$$

So we have the theorem. ■

Lemma 1.1.7

Let $K = F(X)$ be the field of rational functions in X over F. If $u \in K$ with $u \notin F$ let $u = f/g$ where $f, g \in F[X]$ and $\gcd(f, g) = 1$. Let $L = F(u)$. Then

$$[K\colon L] = \max\{\deg(f), \deg(g)\}$$

Proof: By Theorem 1.1.4 we need to find the minimal polynomial of X over L to determine $[K\colon L]$. Consider the polynomial $p(Y) = u \cdot g(Y) - f(Y) \in L[Y]$. Then X is a root of p. Therefore, X is algebraic over L and so $[K\colon L] < \infty$ and $K = L(X)$.

Notice that $\deg(p) = \max\{\deg(f), \deg(g)\}$. Let $f = \sum_{i=0}^{n} a_i X^i$ and $g = \sum_{j=0}^{m} b_i \cdot X^i$. We will show that p is irreducible over L. Now the element u cannot be algebraic over k otherwise by Theorem 1.1.6 we have $[K\colon F] = [K\colon L] \cdot [L\colon F] < \infty$, and we know $F(X)$ is an infinite extension of F. So u is transcendental over F and hence $F[u] \cong F[X]$. Now observe that the polynomial $p \in F[Y][u] \subseteq F(Y)[u]$ which is of degree 1. Hence, p is irreducible over $k(Y)$. Now $p \in k[Y][u] = k[u][Y]$ and therefore p is irreducible over $k(u)$. Hence, p is the minimal polynomial of X over F. So we have the lemma. ■

Note:-

In the proof of Theorem 1.1.6 the proof do not assume that the field extensions needs to be finite.

We will now prove a converse theorem of Lemma 1.1.5.

Theorem 1.1.8

Let K/F be a field extension. If $\alpha_1, \ldots, \alpha_n \in K$ are algebraic over F, then

$$[F(\alpha_1, \ldots, \alpha_n)\colon F] \;\leq\; \prod_{i=1}^{n} [F(\alpha_i)\colon F].$$

Proof: We will prove this by induction on n. The base case, $n = 1$ follows from the Theorem 1.1.6. So suppose this is true for $n - 1$. Take $L = F(\alpha_1, \ldots, \alpha_{n-1})$. Then by induction hypothesis $[L : F] \leq \prod_{i=1}^{n-1} [F(\alpha_i) : F]$. Since α_n is algebraic over F and hence over L, with $\min(L, \alpha_n) \mid \min(F, \alpha_n)$ in $L[X]$, we have $[L(\alpha_n) : L] \leq [F(\alpha_n) : F]$. Therefore, $[F(\alpha_1, \ldots, \alpha_n)\colon L] \leq [F(\alpha_n)\colon F]$. So by Theorem 1.1.6 we have

$$[F(\alpha_1, \ldots, \alpha_n) : F] = [L(\alpha_n) : L] \cdot [L : F] \leq [F(\alpha_n) : F] \cdot \prod_{i=1}^{n-1} [F(\alpha_i) : F].$$

So we have the theorem. ■

The inequality above can be strict in some cases. For example let $a = \sqrt[4]{2}$ and $b = \sqrt[4]{18}$. Then both $[\mathbb{Q}(a)\colon \mathbb{Q}] = [\mathbb{Q}(b)\colon \mathbb{Q}] = 4$ as $X^4 - 2$ and $X^4 - 18$ are both irreducible polynomials over $\mathbb{Q}$. But $[\mathbb{Q}(\sqrt[4]{2}, \sqrt{3})\colon \mathbb{Q}] = 8$ and $\sqrt[4]{18} \in \mathbb{Q}(\sqrt[4]{2}, \sqrt{3})$. Hence, in this the inequality doesn't follow.

So from Theorem 1.1.4 and Theorem 1.1.8 we have the following criterion for an element being algebraic over a field.

Corollary 1.1.9

Let K/F be a field extension and $\alpha \in K$. Then α is algebraic over F if and only if $[F(\alpha) : F] < \infty$. Moreover, K is algebraic over F if $[K : F] < \infty$.

Now we can extend Theorem 1.1.8 to the case of fields generated by arbitrary number of elements or extending field.

Theorem 1.1.10

Let K be a field extension of F and $X \subseteq K$ such that each element of S is algebraic over F. Then $F(S)/F$ is algebraic. If $|S| < \infty$, then $[F(S) : F] < \infty$.

Proof: For any $\alpha \in F(S)$. Theorem 1.1.3 gives $\alpha \in F(a_1, \ldots, a_m)$ for some $a_1, \ldots, a_m \in S$. By Theorem 1.1.8, $F(a_1, \ldots, a_m)$ is algebraic over F. So α is algebraic over F by Theorem 1.1.5. Since α is any arbitrary element of $F(S)$, $F(S)$ is algebraic over F. If S is finite, then $[F(S) : F] < \infty$ by Theorem 1.1.8. ■

We will now show that the property of being algebraic is a transitive property. In the finite extensions case this property easily follows from Theorem 1.1.6 and Corollary 1.1.9. We will show this for general extensions below.

Theorem 1.1.11 Transitivity of Algebraic Extensions

Let $F \subseteq L \subseteq K$ be fields. If L/F and K/L are algebraic, then K/F is algebraic.

Proof: Let $\alpha \in K$. Since K/L is algebraic, let $f(X) = \sum_{i=0}^{d} a_i X^i$ where each $a_i \in L$ is the minimal polynomial of α over L. Since L/F is algebraic and $f \in L[X]$, each a_i is algebraic over F. Consider the field $L' = F(a_0, \ldots, a_d)$. So by Theorem 1.1.8 the extension L'/F is finite i.e., $[L' : F] < \infty$. Since $f \in L'[X]$ and $p(\alpha) = 0$, the element α is algebraic over L' and $[L'(\alpha) : L'] < \infty$. By the tower law,

$$[L'(\alpha) : F] = [L'(\alpha) : L'] \cdot [L' : F] < \infty,$$

Since $F(\alpha) \subseteq L'(\alpha)$, α is algebraic over F. Since α is any arbitrary element of K, K/F is algebraic. ■

Definition 1.1.4: Algebraic Closure in K

Let K be a field extension of F. The set

$$\{\, a \in K : a \text{ is algebraic over } F \,\}$$

is called the algebraic closure of F in K.

Corollary 1.1.12

Let K be a field extension of F and let L be the algebraic closure of F in K. Then L is a field, and therefore is the largest algebraic extension of F contained in K.

Proof: Let $a, b \in L$. Then both are algebraic over F by Theorem 1.1.8 and hence $F(a, b)$ is algebraic over F and $F(a, b) \subseteq L$. Since $a \pm b, a \cdot b, a/b$ (if $b \neq 0$) all are in $F(a, b) \subseteq L$, L is closed under the field operation. So L is a subfield of K. Since each element of K which is algebraic over F is contained in L, L is the largest algebraic extension of F contained in K. ■

1.1.4 Splitting Field

Definition 1.1.5: Splits over K

If K is an extension field of F and if $f(X) \in F[X]$, then f splits over K *if $f(X) = a\prod_i(X-\alpha_i) \in K[X]$ for some $\alpha_1, \ldots, \alpha_n \in K$ and $a \in F$. In other words, f splits over K if f factors completely into linear factors in $K[X]$.*

In order to talk about roots of a given polynomial, we need to have extension fields that contain the roots of the polynomial. The next theorem shows that for any $f \in F[X]$, there is a finite extension of F over which f splits.

Theorem 1.1.13

Let $f(x) \in F[x]$ have degree n. There is an extension field K of F with $[K : F] \leq n$ such that K contains a root of f. In addition, there is a field L containing F with $[L : F] \leq n!$ such that f splits over L.

Proof: Let $p(X)$ is an irreducible factor of $f(X)$ in $F[X]$ and let $K = F[X] / \langle p(X) \rangle$. Thus, K contains a root of $p(X)$ say α. Therefore, α is also a root of $f(X)$. Since $[K\colon F] = \deg(p) \leq n$ we have the first part.

For the second part we will use induction on n. From the first part now we have a field $K \supseteq F$ such that K contains a root $\alpha of F$. So f factors into $f(X) = (X-\alpha)g(X)$ in $K[X]$ where $g(X) \in K[X]$. Now $\deg(g) = n-1$. So by induction hypothesis there exists a finite extension L of K such that $[L\colon K] \leq (n-1)!$ and g splits over L. That means f splits over L and $[L\colon F] = [L\colon K] \cdot [K\colon F] \leq n!$. So we have the theorem. ■

Definition 1.1.6: Splitting Field

Let K be an extension of F and let $f \in F[X]$.

(i) K is called the splitting field of f over F *if and only if f splits over K and $K = F(\alpha_1, \ldots, \alpha_n)$ where $\alpha_1, \ldots, \alpha_n$ are exactly the roots of f.*

(ii) If S is a set of non-constant polynomials over F then K is the splitting field of S over F *if and and only if each $f \in S$ splits over K and $K = F(S_R)$ where S_R is the set of all roots of all $f \in S$.*

So Theorem 1.1.13 yields immediately the existence of splitting fields for a finite set of polynomials. We have proved the existence of splitting fields for finite sets of polynomials. What about infinite sets? Suppose that K is a splitting field over F of the set of all non-constant polynomials over F. We do not know yet that such a field exists or not.

Lemma 1.1.14

If K is a field, then the following statements are equivalent:

(i) There are no algebraic extensions of K other than K itself.

(ii) There are no finite extensions of K other than K itself.

(iii) If L is a field extension of K, then $K = \{a \in L : a \text{ is algebraic over } K\}$.

(iv) Every $f(x) \in K[x]$ splits over K.

(v) Every $f(x) \in K[x]$ has a root in K.

(vi) Every irreducible polynomial over K has degree 1.

Proof: We will show the implications in a circular order, (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (iv) $\Longrightarrow$ (v) $\Longrightarrow$ (vi) $\Longrightarrow$ (i).

(i) $\implies$ (ii): This is clear since any finite extension is algebraic.

(ii) $\implies$ (iii): Let $a \in L$ is algebraic over K. Then $K(a)$ is a finite extension of K. But there are no finite extensions of K other than itself. So $K(a) = K$. Thus, $a \in K$.

(iii) $\implies$ (iv): Let $f \in K[X]$ and L be the splitting field of f over K. Since L is algebraic over K we have $L = K$ i.e., f splits over K.

(iv) $\implies$ (v): This is obviously true.

(v) $\implies$ (vi): Let $f \in K[X]$ is irreducible. Therefore, f has a root in K, so f has a linear factor. But f is irreducible. Hence, $\deg(f) = 1$.

(vi) $\implies$ (i): Let L be an algebraic extension of K. Let $a \in L$ and $p = \min(K, a)$. Then $\deg(p) = 1$. So $[K(a) : K] = 1$ and hence $a \in K$. So $L = K$. ■

Definition 1.1.7: Algebraically Closed Field

If any field K satisfies the equivalent conditions of Lemma 1.1.14 then K is said to be algebraically closed. *If K is an algebraic extension of F then K is said to be* algebraic closure *of F.*

We haven't showed the existence of such theorem or even every field as an algebraic closure. Consider the field $\mathbb{C}$. This field is algebraically closed. But we will not prove if every field has an algebraic closure. We will just state the theorem and move on.

Theorem 1.1.15

Let F is a field. Then F has an algebraic closure.

The existence of an algebraic closure yields immediately the existence of a splitting field for an arbitrary set of non-constant polynomials.

Corollary 1.1.16

Let F be a field and S be a set of non-constant polynomials over F. Then S has a splitting field over F.

Proof: Let K be an algebraic closure of F. Then each $f(x) \in S$ splits over K. Let X be the set of roots of all $f \in S$. Then $F(X) \subseteq K$ is a splitting field for S over F, since each f splits over $F(X)$ and this field is generated by the roots of all the polynomials from S. ■

So using the corollary we can immediately relate algebraic closure with splitting fields.

Corollary 1.1.17

If F is a field then the splitting field of the set of all non-constant polynomials over F is an algebraic closure of F.

But now comes another question. Can one field have multiple algebraic closures? Or even can we have more than one splitting fields? The answer is splitting fields are unique, and hence algebraic closures are also unique. Below we

share the theorem which proves the above. We will not prove it because its beyond our scope of study. But for further reading and understanding the reader can look [Mor96]

Theorem 1.1.18 Isomorphism Extension Theorem, [Mor96, Chapter 1]

Let $\sigma : F \longrightarrow F'$ be a field isomorphism. Let $S = \{f_i(x)\}$ be a set of polynomials over F, and let $S' = \{\sigma(f_i)\}$ be the corresponding set over F'. Let K be a splitting field for S over F, and let K' be a splitting field for S' over F'. Then there is an isomorphism $\tau : K \longrightarrow K'$ with $\tau|_F = \sigma$. Furthermore, if $\alpha \in K$ and α' is any root of $\sigma(\min(F, \alpha))$ in K', then τ can be chosen so that $\tau(\alpha) = \alpha'$.

The isomorphism extension theorem immediately settles the question of uniqueness of splitting fields and algebraic closures: any two splitting fields of the same set of polynomials over F must be isomorphic, since the identity on F extends to an isomorphism between them. As a special case, algebraic closures are unique up to F-isomorphism.

Corollary 1.1.19

Let F be a field, and let S be a subset of $F[x]$. Any two splitting fields of S over F are F-isomorphic. In particular, any two algebraic closures of F are F-isomorphic.

Proof: For the proof of the first statement, the isomorphism extension theorem gives an isomorphism extending id on F between any two splitting fields of S. The second statement follows from the first, since any algebraic closure of F is a splitting field of the set of all non-constant polynomials in $F[x]$. ■

As a corollary to the existence and uniqueness of algebraic closures, we can prove that any algebraic extension of a field F can be viewed as living inside a fixed algebraic closure of F.

Corollary 1.1.20

Let F be a field, and let N be an algebraic closure of F. If K is an algebraic extension of F, then K is isomorphic to a subfield of N.

Proof: Let M be an algebraic closure of K. By Theorem 1.1.11, M is algebraic over F; hence, M is also an algebraic closure of F. Therefore, by the previous corollary, $M \cong N$. If $f : M \longrightarrow N$ is an F-isomorphism, then $f(K)$ is a subfield of N isomorphic to K. ■

§ 1.2 Some Properties of Finite Fields

§ 1.3 Trace and Norm

Chapter 2

Bibliography

[Mor96] Patrick Morandi. *Field and Galois Theory*, volume 167 of *Graduate Texts in Mathematics*. Springer, 1996. doi:10.1007/978-1-4612-4040-2.